CollegeBoard

AP® Physics C: Mechanics Practice Exam

From the 2016 Administration

ADVANCED PLACEMENT PHYSICS C TABLE OF INFORMATION

CONSTANTS AND CONVERSION FACTORS

Proton mass, $m_p = 1.67 \times 10^{-27}$ kg	Electron charge magnitude, $e = 1.60 \times 10^{-19}$ C
Neutron mass, $m_n = 1.67 \times 10^{-27}$ kg	1 electron volt, $1\text{ eV} = 1.60 \times 10^{-19}$ J
Electron mass, $m_e = 9.11 \times 10^{-31}$ kg	Speed of light, $c = 3.00 \times 10^{8}$ m/s
Avogadro's number, $N_0 = 6.02 \times 10^{23}\ \text{mol}^{-1}$	Universal gravitational constant, $G = 6.67 \times 10^{-11}\ (\text{N}\cdot\text{m}^2)/\text{kg}^2$
Universal gas constant, $R = 8.31\ \text{J}/(\text{mol}\cdot\text{K})$	Acceleration due to gravity at Earth's surface, $g = 9.8\ \text{m}/\text{s}^2$
Boltzmann's constant, $k_B = 1.38 \times 10^{-23}$ J/K	

1 unified atomic mass unit,	$1\text{ u} = 1.66 \times 10^{-27}\text{ kg} = 931\ \text{MeV}/c^2$
Planck's constant,	$h = 6.63 \times 10^{-34}\ \text{J}\cdot\text{s} = 4.14 \times 10^{-15}\ \text{eV}\cdot\text{s}$
	$hc = 1.99 \times 10^{-25}\ \text{J}\cdot\text{m} = 1.24 \times 10^{3}\ \text{eV}\cdot\text{nm}$
Vacuum permittivity,	$\varepsilon_0 = 8.85 \times 10^{-12}\ \text{C}^2/(\text{N}\cdot\text{m}^2)$
Coulomb's law constant,	$k = 1/(4\pi\varepsilon_0) = 9.0 \times 10^{9}\ (\text{N}\cdot\text{m}^2)/\text{C}^2$
Vacuum permeability,	$\mu_0 = 4\pi \times 10^{-7}\ (\text{T}\cdot\text{m})/\text{A}$
Magnetic constant,	$k' = \mu_0/(4\pi) = 1 \times 10^{-7}\ (\text{T}\cdot\text{m})/\text{A}$
1 atmosphere pressure,	$1\text{ atm} = 1.0 \times 10^{5}\ \text{N}/\text{m}^2 = 1.0 \times 10^{5}$ Pa

UNIT SYMBOLS								
	meter,	m	mole,	mol	watt,	W	farad,	F
	kilogram,	kg	hertz,	Hz	coulomb,	C	tesla,	T
	second,	s	newton,	N	volt,	V	degree Celsius,	°C
	ampere,	A	pascal,	Pa	ohm,	Ω	electron volt,	eV
	kelvin,	K	joule,	J	henry,	H		

PREFIXES

Factor	Prefix	Symbol
10^{9}	giga	G
10^{6}	mega	M
10^{3}	kilo	k
10^{-2}	centi	c
10^{-3}	milli	m
10^{-6}	micro	μ
10^{-9}	nano	n
10^{-12}	pico	p

VALUES OF TRIGONOMETRIC FUNCTIONS FOR COMMON ANGLES

θ	0°	30°	37°	45°	53°	60°	90°
$\sin\theta$	0	$1/2$	$3/5$	$\sqrt{2}/2$	$4/5$	$\sqrt{3}/2$	1
$\cos\theta$	1	$\sqrt{3}/2$	$4/5$	$\sqrt{2}/2$	$3/5$	$1/2$	0
$\tan\theta$	0	$\sqrt{3}/3$	$3/4$	1	$4/3$	$\sqrt{3}$	∞

The following assumptions are used in this exam.

I. The frame of reference of any problem is inertial unless otherwise stated.
II. The direction of current is the direction in which positive charges would drift.
III. The electric potential is zero at an infinite distance from an isolated point charge.
IV. All batteries and meters are ideal unless otherwise stated.
V. Edge effects for the electric field of a parallel plate capacitor are negligible unless otherwise stated.

ADVANCED PLACEMENT PHYSICS C EQUATIONS

MECHANICS

$v_x = v_{x0} + a_x t$

$x = x_0 + v_{x0} t + \frac{1}{2} a_x t^2$

$v_x^2 = v_{x0}^{\ 2} + 2a_x (x - x_0)$

$\vec{a} = \frac{\sum \vec{F}}{m} = \frac{\vec{F}_{net}}{m}$

$\vec{F} = \frac{d\vec{p}}{dt}$

$\vec{J} = \int \vec{F}\, dt = \bullet \vec{p}$

$\vec{p} = m\vec{v}$

$\left|\vec{F}_f\right| \le \mu \left|\vec{F}_N\right|$

$\bullet E = W = \int \vec{F} \bullet d\vec{r}$

$K = \frac{1}{2} m v^2$

$P = \frac{dE}{dt}$

$P = \vec{F} \bullet \vec{v}$

$\bullet U_g = mg \bullet h$

$a_c = \frac{v^2}{r} = \omega^2 r$

$\vec{\tau} = \vec{r} \times \vec{F}$

$\vec{\alpha} = \frac{\sum \vec{\tau}}{I} = \frac{\vec{\tau}_{net}}{I}$

$I = \int r^2 dm = \sum m r^2$

$x_{cm} = \frac{\sum m_i x_i}{\sum m_i}$

$v = r\omega$

$\vec{L} = \vec{r} \times \vec{p} = I\vec{\omega}$

$K = \frac{1}{2} I \omega^2$

$\omega = \omega_0 + \alpha t$

$\theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha t^2$

a = acceleration
E = energy
F = force
f = frequency
h = height
I = rotational inertia
J = impulse
K = kinetic energy
k = spring constant
ℓ = length
L = angular momentum
m = mass
P = power
p = momentum
r = radius or distance
T = period
t = time
U = potential energy
v = velocity or speed
W = work done on a system
x = position
μ = coefficient of friction
θ = angle
τ = torque
ω = angular speed
α = angular acceleration
ϕ = phase angle

$\vec{F}_s = -k \bullet \vec{x}$

$U_s = \frac{1}{2} k (\bullet x)^2$

$x = x_{\max} \cos(\omega t + \phi)$

$T = \frac{2\pi}{\omega} = \frac{1}{f}$

$T_s = 2\pi \sqrt{\frac{m}{k}}$

$T_p = 2\pi \sqrt{\frac{\ell}{g}}$

$\left|\vec{F}_G\right| = \frac{G m_1 m_2}{r^2}$

$U_G = -\frac{G m_1 m_2}{r}$

ELECTRICITY AND MAGNETISM

$\left|\vec{F}_E\right| = \frac{1}{4\pi\varepsilon_0} \left|\frac{q_1 q_2}{r^2}\right|$

$\vec{E} = \frac{\vec{F}_E}{q}$

$\oint \vec{E} \bullet d\vec{A} = \frac{Q}{\varepsilon_0}$

$E_x = -\frac{dV}{dx}$

$\bullet V = -\int \vec{E} \bullet d\vec{r}$

$V = \frac{1}{4\pi\varepsilon_0} \sum_i \frac{q_i}{r_i}$

$U_E = qV = \frac{1}{4\pi\varepsilon_0} \frac{q_1 q_2}{r}$

$\bullet V = \frac{Q}{C}$

$C = \frac{\kappa \varepsilon_0 A}{d}$

$C_p = \sum_i C_i$

$\frac{1}{C_s} = \sum_i \frac{1}{C_i}$

$I = \frac{dQ}{dt}$

$U_C = \frac{1}{2} Q \bullet V = \frac{1}{2} C (\bullet V)^2$

$R = \frac{\rho \ell}{A}$

$\vec{E} = \rho \vec{J}$

$I = N e v_d A$

$I = \frac{\bullet V}{R}$

$R_s = \sum_i R_i$

$\frac{1}{R_p} = \sum_i \frac{1}{R_i}$

$P = I \bullet V$

A = area
B = magnetic field
C = capacitance
d = distance
E = electric field
ε = emf
F = force
I = current
J = current density
L = inductance
ℓ = length
n = number of loops of wire per unit length
N = number of charge carriers per unit volume
P = power
Q = charge
q = point charge
R = resistance
r = radius or distance
t = time
U = potential or stored energy
V = electric potential
v = velocity or speed
ρ = resistivity
$\bullet$ = flux
κ = dielectric constant

$\vec{F}_M = q\vec{v} \times \vec{B}$

$\oint \vec{B} \bullet d\vec{\ell} = \mu_0 I$

$d\vec{B} = \frac{\mu_0}{4\pi} \frac{I\, d\vec{\ell} \times \hat{r}}{r^2}$

$\vec{F} = \int I\, d\vec{\ell} \times \vec{B}$

$B_s = \mu_0 n I$

$\bullet_B = \int \vec{B} \circ d\vec{A}$

$\varepsilon = \oint \vec{E} \bullet d\vec{\ell} = -\frac{d\bullet_B}{dt}$

$\varepsilon = -L \frac{dI}{dt}$

$U_L = \frac{1}{2} L I^2$

GEOMETRY AND TRIGONOMETRY

Rectangle

$A = bh$

Triangle

$A = \frac{1}{2}bh$

Circle

$A = \pi r^2$

$C = 2\pi r$

$s = r\theta$

Rectangular Solid

$V = \ell wh$

Cylinder

$V = \pi r^2 \ell$

$S = 2\pi r\ell + 2\pi r^2$

Sphere

$V = \frac{4}{3}\pi r^3$

$S = 4\pi r^2$

Right Triangle

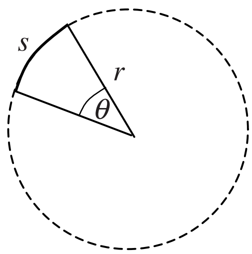
$a^2 + b^2 = c^2$

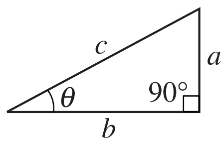
$\sin\theta = \frac{a}{c}$

$\cos\theta = \frac{b}{c}$

$\tan\theta = \frac{a}{b}$

A = area
C = circumference
V = volume
S = surface area
b = base
h = height
ℓ = length
w = width
r = radius
s = arc length
θ = angle





CALCULUS

$$\frac{df}{dx} = \frac{df}{du}\frac{du}{dx}$$

$$\frac{d}{dx}\left(x^n\right) = nx^{n-1}$$

$$\frac{d}{dx}\left(e^{ax}\right) = ae^{ax}$$

$$\frac{d}{dx}(\ln ax) = \frac{1}{x}$$

$$\frac{d}{dx}[\sin(ax)] = a\cos(ax)$$

$$\frac{d}{dx}[\cos(ax)] = -a\sin(ax)$$

$$\int x^n\,dx = \frac{1}{n+1}x^{n+1},\ n \neq -1$$

$$\int e^{ax}\,dx = \frac{1}{a}e^{ax}$$

$$\int \frac{dx}{x+a} = \ln|x+a|$$

$$\int \cos(ax)\,dx = \frac{1}{a}\sin(ax)$$

$$\int \sin(ax)\,dx = -\frac{1}{a}\cos(ax)$$

VECTOR PRODUCTS

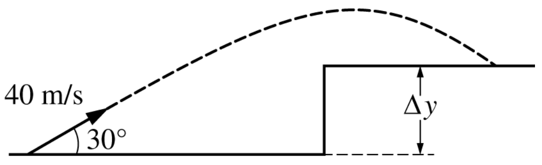
$\vec{A}\bullet\vec{B} = AB\cos\theta$

$\left|\vec{A}\times\vec{B}\right| = AB\sin\theta$

PHYSICS C: MECHANICS
SECTION I
Time—45 minutes
35 Questions

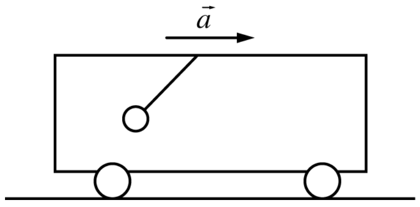
Directions: Each of the questions or incomplete statements below is followed by five suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

Note: To simplify calculations, you may use $g = 10\,\text{m/s}^2$ in all problems.



1. A projectile is launched with a speed of $40\ \text{m/s}$ at an angle of $30°$ above the horizontal, as shown in the figure above. The projectile lands on a plateau 3 s later. The height Δy of the plateau is most nearly

   (A) 15 m
   (B) 30 m
   (C) 45 m
   (D) 60 m
   (E) 75 m



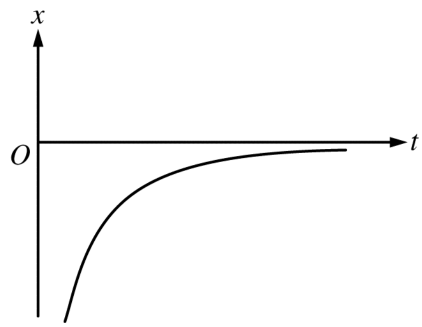
2. A small sphere hangs from a string attached to the ceiling of a uniformly accelerating train car. It is observed that the string makes an angle of $37°$ with respect to the vertical. The magnitude of the acceleration a of the train car is most nearly

   (A) $6.0\ \text{m/s}^2$
   (B) $7.5\ \text{m/s}^2$
   (C) $8.0\ \text{m/s}^2$
   (D) $10\ \text{m/s}^2$
   (E) $13\ \text{m/s}^2$

GO ON TO THE NEXT PAGE.

3. Which of the following statements must be true for a falling object that has been dropped from rest near the surface of Earth?

   (A) The derivative of the distance the object falls with respect to time equals $9.8\ \text{m/s}^2$.
   (B) The object falls a vertical distance of 9.8 m during the first second only.
   (C) The object falls a vertical distance of 9.8 m during each second.
   (D) The speed of the object as it falls is a constant $9.8\ \text{m/s}$.
   (E) The speed of the object increases by $9.8\ \text{m/s}$ during each second.



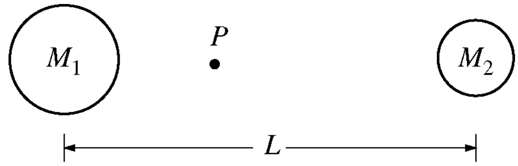
4. The position x as a function of time t for an object moving in a straight line is shown in the graph above. Which of the following best describes the object's speed and direction of motion during the time interval shown?

	Speed	Direction of Motion
(A)	Decreasing	Positive
(B)	Increasing	Positive
(C)	Constant	Positive
(D)	Decreasing	Negative
(E)	Increasing	Negative

GO ON TO THE NEXT PAGE.

5. The force F exerted on a ball during a collision with a wall is given as a function of time t by the equation $F(t) = \alpha t - \beta t^2$, where $\alpha = 400$ N/s and $\beta = 4000$ N/s^2. The ball first contacts the wall at $t = 0$, and the collision lasts for 0.10 s. What is the magnitude of the change in momentum of the ball?

(A) 0
(B) 0.67 kg•m/s
(C) 3.33 kg•m/s
(D) 3.60 kg•m/s
(E) The change in momentum of the ball cannot be determined without knowing the mass of the ball.



6. Two spheres of uniform density are located a distance L apart, as shown in the figure above. The left sphere has a mass M_1, and the right sphere has a mass M_2. The center of mass of the two spheres is labeled point P. Which of the following is a correct expression for the distance from point P to the center of the left sphere?

(A) $\dfrac{M_1}{M_2}L$

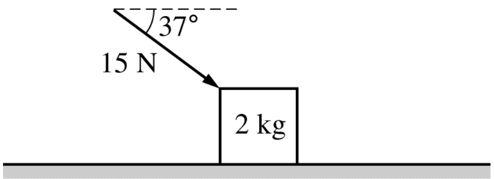
(B) $\dfrac{(M_1 + M_2)}{M_1}L$

(C) $\dfrac{(M_1 + M_2)}{M_2}L$

(D) $\dfrac{M_2}{(M_1 + M_2)}L$

(E) $\dfrac{M_1}{(M_1 + M_2)}L$

Questions 7-8



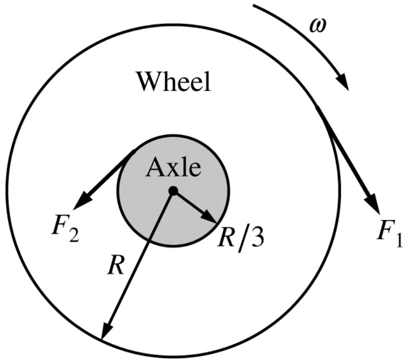
A 2 kg block is initially at rest on a horizontal frictionless table. A force of 15 N is then exerted on the block at an angle of $37°$ to the horizontal, as shown above.

7. The change in the kinetic energy of the block after moving a distance of 3 m is most nearly

   (A) 60 J
   (B) 45 J
   (C) 36 J
   (D) 27 J
   (E) 24 J

8. The magnitude of the force exerted on the block by the table is most nearly

   (A) 35 N
   (B) 32 N
   (C) 29 N
   (D) 20 N
   (E) 11 N

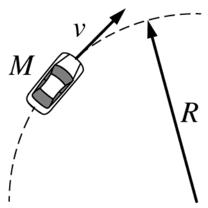


Note: Vectors not drawn to scale.

9. A wheel of radius R is fixed to an axle of radius $R/3$ and rotates at constant angular speed ω, as shown in the figure above. A force of magnitude F_1 is applied tangent to the outer edge of the wheel. A second force of magnitude F_2 is applied tangent to the edge of the axle to keep the wheel rotating at constant angular speed ω. The magnitude F_2 is equal to

   (A) $F_1/9$
   (B) $F_1/3$
   (C) F_1
   (D) $3F_1$
   (E) $9F_1$

Questions 10-11



On a level horizontal table, a toy race car of mass M moves with constant speed v around a flat circular racetrack of radius R.

10. Which of the following best represents the minimum coefficient of static friction required for the race car to continue to follow the circular path shown?

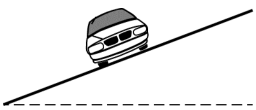
(A) $\dfrac{Rg}{v^2}$

(B) $\dfrac{Mv^2}{R}$

(C) $\dfrac{v^2}{g}$

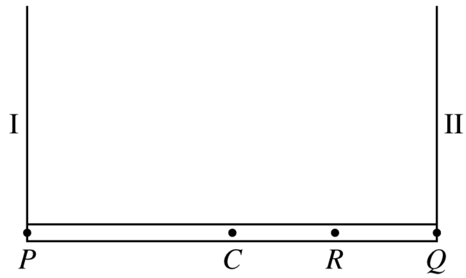
(D) $\dfrac{v^2}{Rg}$

(E) $\dfrac{v^2}{2Rg}$



11. The car is now going around a banked curve, as shown in the figure above. The banked curve is higher toward the outside of the circular turn. The car can now travel at a higher speed without sliding because the banked curve

(A) increases the coefficient of friction between the car and the track.
(B) increases the momentum of the car.
(C) allows a component of the normal force to point toward the center of the track.
(D) allows a component of the weight of the car to point toward the center of the track.
(E) allows gravity to do work on the car.



12. A horizontal uniform plank is supported by ropes I and II at points P and Q, respectively, as shown above. The two ropes have negligible mass. The tension in rope I is 150 N. The point at which rope II is attached to the plank is now moved to point R halfway between point Q and point C, the center of the plank. The plank remains horizontal. Which of the following are most nearly the new tensions in the two ropes?

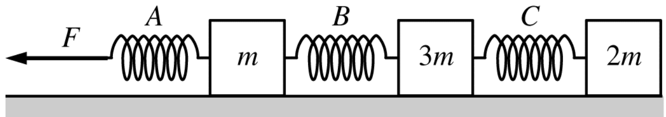
	Tension in I	Tension in II
(A)	75 N	225 N
(B)	100 N	200 N
(C)	112.5 N	112.5 N
(D)	112.5 N	187.5 N
(E)	150 N	300 N

13. The potential energy U as a function of the position x of an object is given by $U(x) = -400/x$, where U is in joules and x is in meters. The object is released from rest at position $x = 30$ m and is free to move along the x-axis. When the object has moved a distance of 10 m from its initial position, the magnitude of the force associated with this potential energy function is

(A) 20 N
(B) 10 N
(C) 4.0 N
(D) 1.0 N
(E) 0.25 N

14. A student pushes a box across a rough horizontal floor. If the amount of work done by the student on the box is 100 J and the amount of energy dissipated by friction is 40 J, what is the change in kinetic energy of the box?

(A) 0 J
(B) 40 J
(C) 60 J
(D) 100 J
(E) 140 J



15. Three blocks of masses m, $3m$, and $2m$ resting on a frictionless horizontal surface are connected to identical ideal springs, as shown above. A force of magnitude F directed to the left is then applied to the left end of spring A. Which spring is stretched the most when the blocks are all moving with the same acceleration?

(A) A
(B) B
(C) C
(D) None, because the springs do not stretch.
(E) None, because the springs all stretch the same amount.

GO ON TO THE NEXT PAGE.

16. At time $t = 0$, a ball is thrown with initial speed v_0 at an angle θ above the horizontal. Air resistance is negligible. Which of the following best represents the speed of the ball as a function of time t while the ball is in flight?

(A) $v_0 \sin\theta - gt$

(B) $v_0 \cos\theta - gt$

(C) $\sqrt{v_0^2 \cos^2\theta + (v_0 \sin\theta - gt)^2}$

(D) $\sqrt{(v_0 \cos\theta - gt)^2 + v_0^2 \sin^2\theta}$

(E) $\sqrt{(v_0 \cos\theta - gt)^2 + (v_0 \sin\theta - gt)^2}$

17. The position x of an object is given as a function of time t by the equation $x = 8 + 4t - 6t^3$, where x is in meters and t is in seconds. What is the maximum positive velocity attained by this object?

(A) 4 m/s
(B) 8 m/s
(C) 18 m/s
(D) 36 m/s
(E) There is no maximum positive velocity because the object never moves in the positive direction.

18. Two balls, A and B, have the same mass and diameter but are made from different types of rubber. The balls are dropped from the same height above the floor. After colliding with the floor, ball A bounces higher than ball B bounces. Which of the following quantities must necessarily be larger for ball A than for ball B ?

(A) The average force exerted by the floor
(B) The amount of time in contact with the floor
(C) The impulse exerted by the floor
(D) The momentum just before colliding with the floor
(E) The kinetic energy just before colliding with the floor

19. Object A of mass M is moving east at speed v. It collides with object B of mass $2M$ that was initially at rest. The motion of the objects before and after the collision is along the same line. After the collision, object A is moving west at a speed of $v/3$. What is the speed of object B immediately after the collision?

(A) $v/3$
(B) $v/2$
(C) $2v/3$
(D) v
(E) $2v$

Questions 20-21

A wheel of mass m, which has a rotational inertia I and a radius r, rotates with an angular speed ω about an axis through its center. A retarding force F is applied tangentially to the rim of the wheel.

20. Which of the following is equal to the magnitude of the angular acceleration of the wheel?

(A) $\dfrac{F\omega}{m}$

(B) $\dfrac{Fr}{I}$

(C) $\dfrac{\omega^2}{r}$

(D) $I\omega$

(E) ωr

21. The retarding force F finally stops the rotation of the wheel. Which of the following best represents the total reduction in mechanical energy in the process of stopping the wheel?

(A) Fr

(B) $I\omega^2$

(C) $\omega^2 r$

(D) Fr^2

(E) $\dfrac{1}{2}I\omega^2$

22. An airplane travels horizontally at a constant velocity $\vec{v}$. An object is dropped from the plane, and one second later another object is dropped from the plane. If air resistance is negligible, what happens to the vertical distance between the two objects while they are both falling?

(A) It increases.
(B) It decreases.
(C) It remains the same.
(D) It depends on the mass of the objects.
(E) It depends on the horizontal speed of the plane.

23. A boat moves across a river. The river's velocity relative to the land is $\vec{v}_{RL}$, and the boat's velocity relative to the river is $\vec{v}_{BR}$. Which of the following best represents the velocity of the boat relative to the land?

(A) $\vec{v}_{BR} + \vec{v}_{RL}$

(B) $\vec{v}_{BR} - \vec{v}_{RL}$

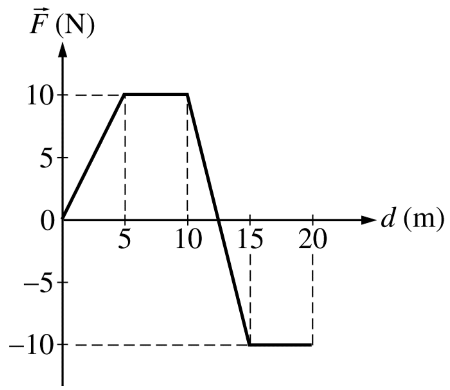
(C) $-\vec{v}_{BR} + \vec{v}_{RL}$

(D) $|\vec{v}_{BR}| + |\vec{v}_{RL}|$

(E) $|\vec{v}_{BR}| - |\vec{v}_{RL}|$

Questions 24-25

The following graph shows the force $\vec{F}$ exerted on a 2 kg object as a function of the distance d that the object travels. The object is at rest at $d = 0$ and travels on a horizontal, frictionless surface along the line of action of the force.



24. The work done on the object by the force $\vec{F}$ during the first 10 m of travel is most nearly

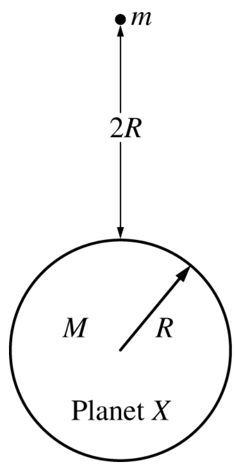
(A) 7.5 J
(B) 10 J
(C) 50 J
(D) 75 J
(E) 100 J

25. The kinetic energy of the 2 kg object when d equals 20 m is the same as when d is most nearly

(A) 0 m
(B) 5 m
(C) 10 m
(D) 12.5 m
(E) 15 m

26. The acceleration due to gravity on the surface of the Moon is about $1/6$ of that on the surface of Earth. An astronaut weighs 600 N on Earth. The astronaut's mass on the Moon is most nearly

(A) 600 kg
(B) 360 kg
(C) 100 kg
(D) 60 kg
(E) 10 kg



27. Planet X has a mass M, radius R, and no atmosphere. An object of mass m is located a distance $2R$ above the surface of planet X, as shown in the figure above. The object is released from rest and falls to the surface of the planet. What is the speed of the object just before it reaches the surface of planet X ?

(A) $\sqrt{\dfrac{2GM}{3R}}$

(B) $\sqrt{\dfrac{4GM}{3R}}$

(C) $\sqrt{\dfrac{8GM}{3R}}$

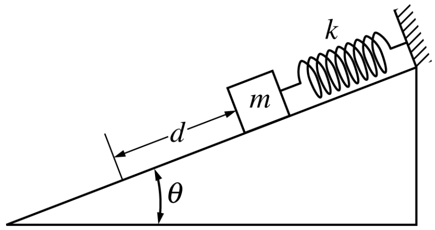
(D) $\sqrt{\dfrac{3GM}{R}}$

(E) $\sqrt{\dfrac{6GM}{R}}$

GO ON TO THE NEXT PAGE.

28. A ball is kicked from the ground with an initial speed v_0 at an angle θ above the horizontal. Which of the following best describes the magnitudes of the velocity and acceleration of the ball when it reaches the highest point of its trajectory?

	Velocity	Acceleration
(A)	0	0
(B)	$v_0 \cos\theta$	0
(C)	$v_0 \sin\theta$	0
(D)	$v_0 \cos\theta$	$9.8\ \text{m/s}^2$
(E)	$v_0 \sin\theta$	$9.8\ \text{m/s}^2$



29. A block of mass m is attached to a spring and placed on an inclined plane. The spring has a spring constant k, and there is negligible friction between the block and inclined plane. The block is released from rest at the position shown in the figure above with the spring initially unstretched. What distance d does the block slide down the plane before coming momentarily to rest?

(A) $\dfrac{2mg}{k}$

(B) $\dfrac{mg}{k}$

(C) $\dfrac{mg\sin\theta}{k}$

(D) $\dfrac{2mg\sin\theta}{k}$

(E) $\sqrt{\dfrac{mg\sin\theta}{k}}$

GO ON TO THE NEXT PAGE.

30. A 5.0 kg object suspended on a spring oscillates such that its position x as a function of time t is given by the equation $x(t) = A\cos(\omega t)$, where $A = 0.80$ m and $\omega = 2.0\ \text{s}^{-1}$. What is the magnitude of the maximum net force on the object during the motion?

(A) 3.2 N
(B) 4.0 N
(C) 8.0 N
(D) 12.8 N
(E) 16.0 N

31. Consider Earth to be stationary, and the Moon as orbiting Earth in a circle of radius R. If the masses of Earth and the Moon are M_E and M_M, respectively, which of the following best represents the total mechanical energy of the Earth-Moon system?

(A) $\dfrac{GM_EM_M}{2R}$

(B) $\dfrac{GM_EM_M}{R}$

(C) $-\dfrac{GM_EM_M}{2R}$

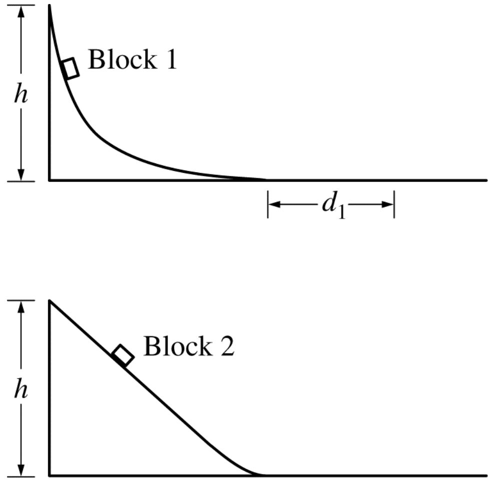
(D) $-\dfrac{GM_EM_M}{R}$

(E) $-\dfrac{2GM_EM_M}{R}$

GO ON TO THE NEXT PAGE.

32. A small rock is launched straight upward from the surface of a planet with no atmosphere. The initial speed of the rock is twice the escape speed v_e of the rock from the planet. If gravitational effects from other objects are negligible, the speed of the rock at a very great distance from the planet will approach a value of

(A) zero
(B) $v_e/2$
(C) v_e
(D) $\sqrt{2}v_e$
(E) $\sqrt{3}v_e$



33. Identical blocks 1 and 2 start from rest at height h and travel down two differently shaped tracks onto identical rough horizontal surfaces, as shown in the figures above. Both tracks are frictionless, and both blocks make a smooth transition onto the rough horizontal surfaces. Block 1 comes to rest at a horizontal distance d_1 from the bottom of the track. Which of the following best describes and explains the horizontal distance d_2 that block 2 travels before coming to rest?

(A) $d_2 < d_1$, because block 2 reaches its maximum speed later than block 1
(B) $d_2 < d_1$, because block 2 has a lower maximum speed than block 1
(C) $d_2 = d_1$, because the blocks have the same mass
(D) $d_2 = d_1$, because the blocks start at the same height
(E) $d_2 > d_1$, because the track for block 2 is shorter

	Mass (kg)	Area $\left(\text{cm}^2\right)$	μ_s
Block 1	0.75	75	0.20
Block 2	3.0	100	0.40

34. In an experiment, two blocks of different roughness are placed on the same rough wooden board. The table above shows three values for each block: the mass, the area of the block in contact with the board, and the coefficient of static friction μ_s between the block and the board. The experiment begins with the board horizontal. One end of the board is then slowly raised. Which of the following correctly identifies the block that will start sliding down the board first and why?

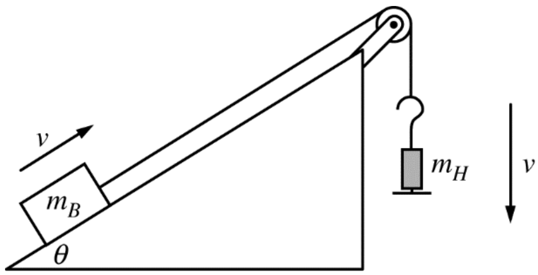
(A) Block 1, because it has less mass per unit area
(B) Block 1, because the coefficient of friction is smaller
(C) Block 2, because it has greater mass
(D) Block 2, because it has a smaller surface area
(E) Block 2, because the coefficient of friction is larger

35. An equation for the motion of an object is given as $2vt - 4x = \quad Bt^2$, where B is a constant. The variable v indicates velocity, in meters per second; t indicates time, in seconds; and x indicates displacement, in meters. What is the unit of measure for B ?

(A) s
(B) m/s
(C) m/s^2
(D) $\text{m}^2\text{/s}$
(E) $\text{m}^2\text{/s}^2$

PHYSICS C: MECHANICS
SECTION II
Time—45 minutes
3 Questions

Directions: Answer all three questions. The suggested time is about 15 minutes for answering each of the questions, which are worth 15 points each. The parts within a question may not have equal weight. Show all your work in this booklet in the spaces provided after each part.



Mech.1.

A block of mass m_B is being pulled up a ramp by a hanging object of mass m_H, as shown in the figure above. The blocks are attached to each other by a light string that passes over a pulley of negligible mass and friction. Both the block and the hanging object are moving at constant speed, and the angle of the ramp with the horizontal is θ. The coefficient of kinetic friction between the block and the ramp is μ_k.

(a) On the dots below, which represent the block and the hanging object, draw and label the forces (not components) that act on the block and the hanging object. Each force must be represented by a distinct arrow starting on, and pointing away from, the appropriate dot.

Block　　　　　　　　　　Hanging Object

(b) Derive an expression for the mass m_H of the hanging object. Express your answer in terms of m_B, θ, μ_k, and physical constants, as appropriate. If you need to draw anything other than what you have shown in part (a) to assist in your solution, use the space below. Do NOT add anything to the figure in part (a).

It is determined that $m_B = 2.0$ kg, $\theta = 30^\circ$, and $\mu_k = 0.10$. While the block is moving up the ramp and the hanging object is moving downward, the string is suddenly cut.

(c)

i. Calculate the magnitude of the acceleration of the block immediately after the string is cut.

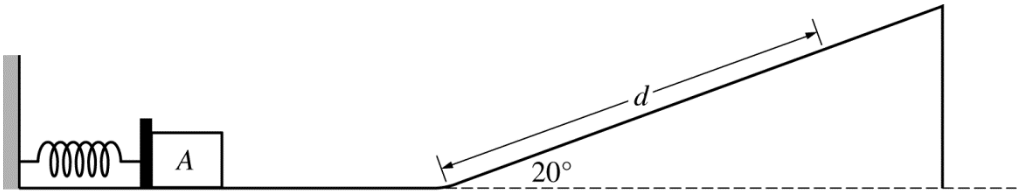
ii. What is the direction of the acceleration of the block immediately after the string is cut?

____ Up the ramp ______ Down the ramp

____ No direction, because the acceleration is zero

iii. Determine the magnitude of the acceleration of the hanging object immediately after the string is cut.

(d) Assume the block is moving up the ramp at 2.0 m/s when the string is cut. Calculate how far the block travels up the ramp from the time the string is cut until the block stops, assuming there is sufficient distance on the ramp.



Note: Figure not drawn to scale.

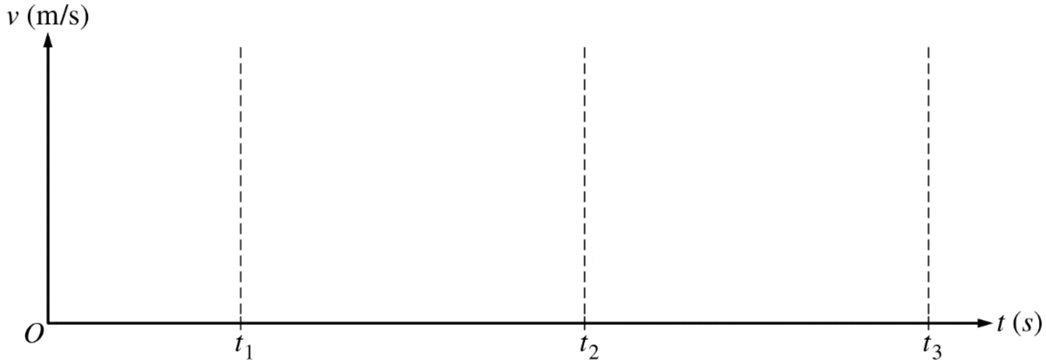
Mech.2.

An experiment is performed on a frictionless surface consisting of a horizontal section and a ramp inclined at a $20°$ angle. Block A of mass 3.0 kg rests on the horizontal section against but not attached to an ideal spring at its equilibrium length, as shown in the figure above. The spring has a spring constant of $k = 700\ \text{N/m}$. Block A is pushed to the left, compressing the spring 0.15 m, and then released.

(a) Calculate the speed of block A when it leaves the spring.

(b) Calculate the maximum distance d block A travels up the ramp.

(c) On the axis below, sketch a graph of the speed v of block A as a function of time t from $t = 0$ to $t = t_3$. Block A is released at time $t = 0$, loses contact with the spring at time $t = t_1$, transitions to the ramp at time $t = t_2$, and reaches its maximum height at time $t = t_3$.



Note: Time intervals not to scale.

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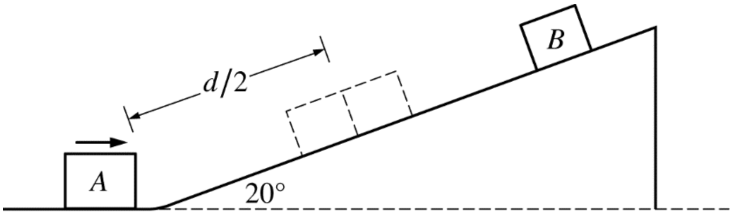


Note: Figure not drawn to scale.

In a second experiment, block *B*, of mass 0.50 kg, is placed on the ramp a distance of *d* from the bottom of the ramp and held in place. Block *A* is released from the spring as in the first experiment. Block *B* is released from rest so that it collides with block *A* as block *A* travels up the ramp. The two blocks collide at a distance of $d/2$ up the ramp. The blocks stick together after the collision.

(d)

i. Calculate the speed of the two blocks immediately after the collision.

ii. In what direction, if any, are the two blocks moving immediately after the collision?

____ Up the ramp ____ Down the ramp

____ Undefined, because the blocks momentarily come to rest after the collision

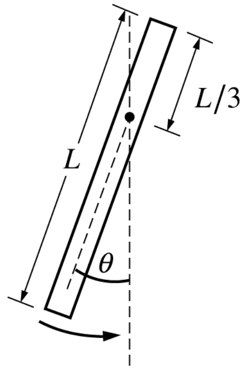
(e)

i. Calculate the magnitude of the impulse exerted on block *A* by block *B*.

ii. In what direction, if any, is the impulse exerted on block *A* by block *B* ?

____ Up the ramp ____ Down the ramp

____ Undefined, because the impulse is zero



Mech.3.

A uniform rod of mass M and length L is pivoted at a point a distance $L/3$ from the top end, as shown above. The rod is pulled back so that it makes a small angle with the vertical and is then released.

(a) Using integral calculus, show that the rotational inertia for the rod around its pivot is $ML^2/9$.

(b)

i. Using Newton's 2nd law in rotational form, write but do NOT solve a differential equation in terms of M, L, and physical constants, as appropriate, that can be used to determine the angular displacement theta as a function of time t.

ii. Using the differential equation from part (b)i, show that the period of oscillation for the rod is given by the expression $T = 2\pi\sqrt{\dfrac{2L}{3g}}$.

An experiment is performed with thin, uniform metal rods of different lengths, each pivoted around a point one third the length of the rod from the top end and used as a physical pendulum, as shown in the figure on the previous page. Each rod is pulled back so that it makes a small angle with the vertical and is then released, and the period of oscillation is measured. The data are recorded in the table below.

Length (m)	0.5	1.0	1.5	2.0	2.5
Period (s)	1.2	1.6	2.0	2.3	2.6

(c) Indicate below which quantities should be graphed to yield a straight line whose slope could be used to calculate a numerical value for the acceleration due to gravity g.

Horizontal axis: ____________________

Vertical axis: ____________________

Use the remaining rows in the table above, as needed, to record any quantities that you indicated that are not given.

(d) Plot the straight line data points on the graph below. Clearly scale and label all axes, including units if appropriate. Draw a straight line that best represents the data.

(e) Using your straight line, determine an experimental value for g.

GO ON TO THE NEXT PAGE.

Answer Key for AP Physics C: Mechanics
Practice Exam, Section I

Question 1: A
Question 2: B
Question 3: E
Question 4: A
Question 5: B
Question 6: D
Question 7: C
Question 8: C
Question 9: D
Question 10: D
Question 11: C
Question 12: B
Question 13: D
Question 14: C
Question 15: A
Question 16: C
Question 17: A
Question 18: C
Question 19: C
Question 20: B
Question 21: E
Question 22: A
Question 23: A
Question 24: D
Question 25: B
Question 26: D
Question 27: B
Question 28: D
Question 29: D
Question 30: E
Question 31: C
Question 32: E
Question 33: D
Question 34: B
Question 35: C

2016 AP Physics C: Mechanics
Question Descriptors and Performance Data

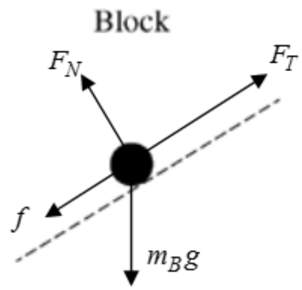
Multiple-Choice Questions

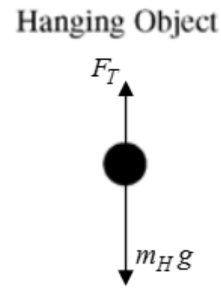
Question	Topic	Key	% Correct
1	Kinematics	A	66
2	Kinematics	B	38
3	Kinematics	E	87
4	Kinematics	A	72
5	System Of Particles & Linear Momentum	B	57
6	System Of Particles & Linear Momentum	D	44
7	Work/Energy/Power	C	77
8	Newton's Laws	C	67
9	Rotation & Circular Motion	D	64
10	Rotation & Circular Motion	D	65
11	Rotation & Circular Motion	C	62
12	Rotation & Circular Motion	B	29
13	Work/Energy/Power	D	14
14	Work/Energy/Power	C	74
15	Oscillations	A	56
16	Kinematics	C	66
17	Kinematics	A	75
18	System Of Particles & Linear Momentum	C	49
19	System Of Particles & Linear Momentum	C	60
20	Rotation & Circular Motion	B	69
21	Rotation & Circular Motion	E	73
22	Kinematics	A	45
23	Kinematics	A	54
24	Work/Energy/Power	D	87
25	Work/Energy/Power	B	63
26	Gravity	D	51
27	Gravity	B	34
28	Kinematics	D	74
29	Oscillations	D	18
30	Oscillations	E	34
31	Gravity	C	24
32	Gravity	E	7
33	Newton's Laws	D	59
34	Newton's Laws	B	59
35	Kinematics	C	77

Question 1

15 points total | **Distribution of points**

(a) 4 points





For correctly drawing and labeling the force of tension and the force of friction for the Block	1 point
For correctly drawing and labeling the normal force vector for the Block	1 point
For correctly drawing and labeling the weight vector for the Block	1 point
For correctly drawing and labeling both vertical vectors for the Hanging Object	1 point

Note: a maximum of three points can only be earned if there are any extraneous vectors

(b) 3 points

For a correct Newton's 2nd law equation of the two-block system — 1 point

$m_H g - F_T = m_H a = 0$
$F_T - m_B g \sin\theta - f = m_B a = 0$ or $m_H g - m_B g \sin\theta - f = (m_H + m_B)a = 0$

For setting the acceleration of the two-block system equal to zero either implicitly or explicitly — 1 point

$m_H g - m_B g \sin\theta - \mu_k m_B g \cos\theta = 0$

For a correct answer — 1 point

$m_H = m_B(\sin\theta + \mu_k \cos\theta)$

(c)
i. 3 points

For correctly applying Newton's second law to the block — 1 point

$-m_B g \sin\theta - \mu_k m_B g \cos\theta = m_B a$

$-g(\sin\theta + \mu_k \cos\theta) = a$

For correctly substituting into the above equation — 1 point

$a = -(9.8\ \text{m/s}^2)((0.10)\cos 30 + \sin 30)$

For a correct answer — 1 point

$a = 5.75\ \text{m/s}^2$ (Note: $a = 5.87\ \text{m/s}^2$ if $g = 10\ \text{m/s}^2$ is used)

Question 1 (continued)

Distribution of points

ii. 1 point

For selecting "Down the ramp" — 1 point

iii. 1 point

For a correct answer — 1 point

$a = g = 9.81 \text{ m/s}^2$ (Note: $a = g = 10 \text{ m/s}^2$ is also acceptable)

Note: No work needs to be shown for this part

(d) 3 points

For using a correct kinematics equation for calculating the distance traveled by the block — 1 point

$v_2^2 = v_1^2 + 2ad$

For setting the final velocity of the block equal to zero — 1 point

$0 = v_1^2 + 2ad$

$$d = \frac{-v_1^2}{2a} = \frac{-(2.0 \text{ m/s})^2}{(2)\left(-5.75 \text{ m/s}^2\right)}$$

For a correct answer — 1 point

$d = 0.348$ m (Note: $d = 0.341$ m if $g = 10 \text{ m/s}^2$ is used)

Alternate solution

For using a correct statement of conservation of energy — *1 point*

$KE_i + W_f = PE_f$

For converting h to d — *1 point*

$h = d\sin\theta$

For a correct answer — *1 point*

$d = 0.348$ m *(Note:* $d = 0.341$ m *if* $g = 10 \text{ m/s}^2$ *is used)*

Question 2

15 points total | **Distribution of points**

(a) 2 points

For correctly using conservation of energy to determine the speed of block A — 1 point

$U_{S1} = K_2$

$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$

$v = \sqrt{\frac{k}{m}}x = \sqrt{\frac{(700\ \text{N/m})}{(3.0\ \text{kg})}}(0.15\ \text{m})$

For a correct answer — 1 point

$v = 2.29\ \text{m/s}$

(b) 2 points

For correctly using conservation of energy to determine the distance block A travels up the ramp — 1 point

$K_1 = U_{g2}$

$\frac{1}{2}mv^2 = mgh = mgd\sin\theta$

$d = \frac{mv^2}{2mg\sin\theta} = \frac{(3.0\ \text{kg})(2.29\ \text{m/s})^2}{(2)(3.0\ \text{kg})(9.8\ \text{m/s}^2)(\sin 20)}$

For a correct answer — 1 point

$d = 0.783\ \text{m}$ (Note: $d = 0.767\ \text{m}$ if $g = 10\ \text{m/s}^2$ is used)

Alternate Solution

For correctly using conservation of energy to determine the distance block A travels up the ramp — *1 point*

$U_{S1} = U_{g2}$

$\frac{1}{2}kx^2 = mgh = mgd\sin\theta$

$d = \frac{kx^2}{2mg\sin\theta} = \frac{(700\ \text{N/m})(0.15\ \text{m})^2}{(2)(3.0\ \text{kg})(9.8\ \text{m/s}^2)(\sin 20)}$

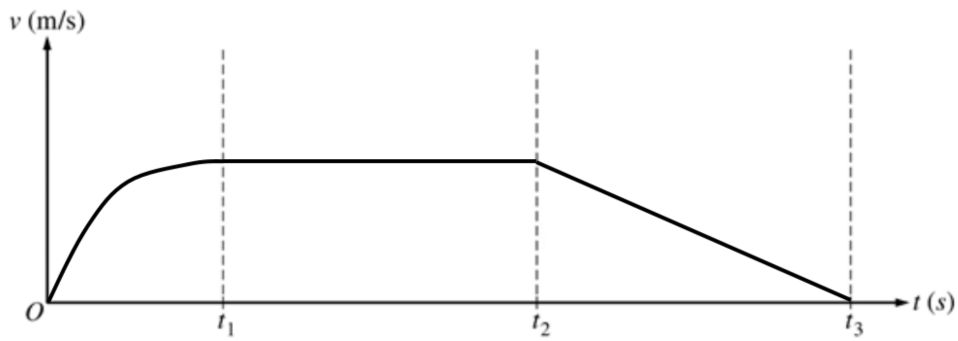
For a correct answer — *1 point*

$d = 0.783\ \text{m}$ (Note: $d = 0.767\ \text{m}$ if $g = 10\ \text{m/s}^2$ is used)

Question 2 (continued)

Distribution of points

(c) 3 points



For a concave down curve that starts at the origin $t = 0$ and becomes horizontal as $t = t_1$	1 point
For a horizontal line from $t = t_1$ to $t = t_2$ that is continuous with first section	1 point
For a straight line with a negative slope from $t = t_2$ to the horizontal axis at $t = t_3$ that is continuous with second section	1 point

Question 2 (continued)

Distribution of points

(d)

i. 5 points

For correctly using conservation of energy to determine speed of block A — 1 point

$K_1 = U_{g2} + K_2$

$\frac{1}{2}mv_1^2 = mgh + \frac{1}{2}mv_2^2$

$\frac{1}{2}mv_1^2 = mg\frac{d}{2}\sin\theta + \frac{1}{2}mv_2^2$

For correctly substituting into the equation above — 1 point

$v_1^2 = gd\sin\theta + v_2^2$

$(2.29\ \text{m/s})^2 = \left(9.8\ \text{m/s}^2\right)(0.783\ \text{m})(\sin 20) + v^2$

For a correct answer — 1 point

$v_A = 1.62\ \text{m/s}$ (Note: $v_A = 1.60\ \text{m/s}$ if $g = 10\ \text{m/s}^2$ is used)

For correctly using conservation of energy to determine speed of block B — 1 point

$U_{g1} = K_2$

$mgh = \frac{1}{2}mv^2$

$mg\frac{d}{2}\sin\theta = \frac{1}{2}mv^2$

$v = \sqrt{\left(9.8\ \text{m/s}^2\right)(0.783)(\sin 20)}$

Correct answer

$v_B = 1.62\ \text{m/s}$ (Note: $v_B = 1.64\ \text{m/s}$ if $g = 10\ \text{m/s}^2$ is used)

For correctly using conservation of momentum to determine the speed of the blocks after the collision — 1 point

$m_A v_A + m_B v_B = (m_A + m_B)v_f$

$$v_f = \frac{(3.0\ \text{kg})(1.62\ \text{m/s}) + (0.50\ \text{kg})(-1.62\ \text{m/s})}{(3.0\ \text{kg} + 0.50\ \text{kg})}$$

Correct answer

$v_f = 1.16\ \text{m/s}$ (Note: $v_f = 1.14\ \text{m/s}$ if $g = 10\ \text{m/s}^2$ is used)

Alternate solution:

For correctly calculating the acceleration of block A — *1 point*

$F = m_A a = m_A g\sin 20^\circ$

$a = g\sin 20^\circ = 3.35\ \text{m/s}^2$

Question 2 (continued)

Distribution of points

For correctly substituting into a correct kinematic equation — *1 point*

$$v_{fA}^2 = v_{0A}^2 + 2a(d/2)$$
$$v_{fA}^2 = (2.29\ \text{m/s})^2 + 2\left(-3.35\ \text{m/s}^2\right)\left(\frac{0.783\ \text{m}}{2}\right)$$

For a correct answer — *1 point*

$v_A = 1.62\ \text{m/s}$ *(Note: $v_A = 1.60\ \text{m/s}$ if $g = 10\ \text{m/s}^2$ is used)*

For using a correct kinematic equation to determine the speed of block B — *1 point*

$$v_{fB}^2 = v_{0B}^2 + 2a(d/2)$$
$$v_{fB}^2 = 0 + 2\left(3.35\ \text{m/s}^2\right)\left(\frac{0.783\ \text{m}}{2}\right)$$
$$v_{fB} = 1.62\ \text{m/s}$$

For correctly using conservation of momentum to determine the speed of the blocks after the collision — *1 point*

$$m_A v_A + m_B v_B = (m_A + m_B)v_f$$
$$v_f = \frac{(3.0\ \text{kg})(1.62\ \text{m/s}) + (0.50\ \text{kg})(-1.62\ \text{m/s})}{(3.0\ \text{kg} + 0.50\ \text{kg})}$$

Correct answer

$v_f = 1.16\ \text{m/s}$ *(Note: $v_f = 1.14\ \text{m/s}$ if $g = 10\ \text{m/s}^2$ is used)*

ii. 1 point

For selecting "Up the ramp" — 1 point

(e)

i. 1 point

Correctly using the impulse equation

$$J = m(v_2 - v_1)$$
$$J = (3.0\ \text{kg})(1.16\ \text{m/s} - 1.62\ \text{m/s})$$

For an answer consistent with the speed of the blocks determined in part (d) — 1 point

$J = -1.38\ \text{N•s}$ (Note: $J = -1.38\ \text{N•s}$ if $g = 10\ \text{m/s}^2$ is used)

Note: The negative sign is not needed since the question asks for magnitude

ii. 1 point

For selecting "Down the ramp" — 1 point

Question 3

15 points total | **Distribution of points**

(a) 3 points

For using integral calculus to attempt to calculate the rotational inertia of the rod — 1 point

$I = \int r^2\,dm$ & $\lambda = M/L$ & $dm = \lambda\,dr$

$I = \int \lambda r^2\,dr$

For correctly integrating the formula above — 1 point

$I = \lambda\left[\dfrac{r^3}{3}\right]$

For integrating using the correct limits or correct constant of integration — 1 point

$$I = \frac{\lambda}{3}\left(\left[r^3\right]_{r=-L/3}^{r=2L/3}\right) = \frac{\lambda}{3}\left[\left(\left(\frac{2L}{3}\right)^3 - \left(-\frac{L}{3}\right)^3\right)\right]$$

$$I = \frac{M}{3L}\left(\frac{8L^3}{27} - \left(-\frac{L^3}{27}\right)\right) = \left(\frac{M}{3L}\right)\left(\frac{9L^3}{27}\right)$$

Correct answer:

$I = ML^2/9$

Alternate Solution

For using integral calculus to attempt to calculate the rotational inertia of the rod about its center — *1 point*

$I = \int r^2\,dm$ & $\lambda = M/L$ & $dm = \lambda\,dr$

$I = \int \lambda r^2\,dr = \lambda\left[\dfrac{r^3}{3}\right]$

For integrating using the correct limits or correct constant of integration — *1 point*

$$I = \frac{\lambda}{3}\left(\left[r^3\right]_{r=-L/2}^{r=L/2}\right) = \frac{\lambda}{3}\left[\left(\left(\frac{L}{2}\right)^3 - \left(-\frac{L}{2}\right)^3\right)\right] = \frac{M}{3L}\left(\frac{L^3}{8} - \left(-\frac{L^3}{8}\right)\right) = \left(\frac{M}{3L}\right)\left(\frac{2L^3}{8}\right)$$

$I = \dfrac{1}{12}ML^2$

For correctly using the Parallel-Axis Theorem — *1 point*

$$I = I_{CM} + Mh^2 = \frac{1}{12}ML^2 + M\left(\frac{L}{6}\right)^2 = \left(\frac{1}{12} + \frac{1}{36}\right)ML^2 = \left(\frac{3+1}{36}\right)ML^2$$

Correct answer:

$I = ML^2/9$

Question 3 (continued)

Distribution of points

(b)

i. 2 points

For using a correct expression of Newton's 2nd law for the rotation of the rod — 1 point

$\tau = Fr_{\perp} = I\alpha$

$-Mg\left(\frac{L}{6}\right)\sin\theta = \frac{ML^2}{9}\alpha$

For expressing the equation as a differential equation — 1 point

$-\frac{3}{2}\frac{g}{L}\sin\theta = \frac{d^2\theta}{dt^2}$

ii. 2 points

For using the small-angle approximation; where $\sin\theta \approx \theta$ — 1 point

$-\frac{3}{2}\frac{g}{L}\theta = \frac{d^2\theta}{dt^2}$

For comparing above equation to the rotational form of simple harmonic motion or for solving the differential equation — 1 point

$-\frac{3}{2}\frac{g}{L}\theta = \frac{d^2\theta}{dt^2} = \alpha = -\theta\omega^2$

$\omega^2 = \left(\frac{2\pi}{T}\right)^2 = \frac{3}{2}\frac{g}{L}$

Correct answer:

$T = 2\pi\sqrt{\frac{2L}{3g}}$

(c) 1 point

For selecting appropriate variable to create a straight line relationship — 1 point

Example:

Horizontal axis: Length

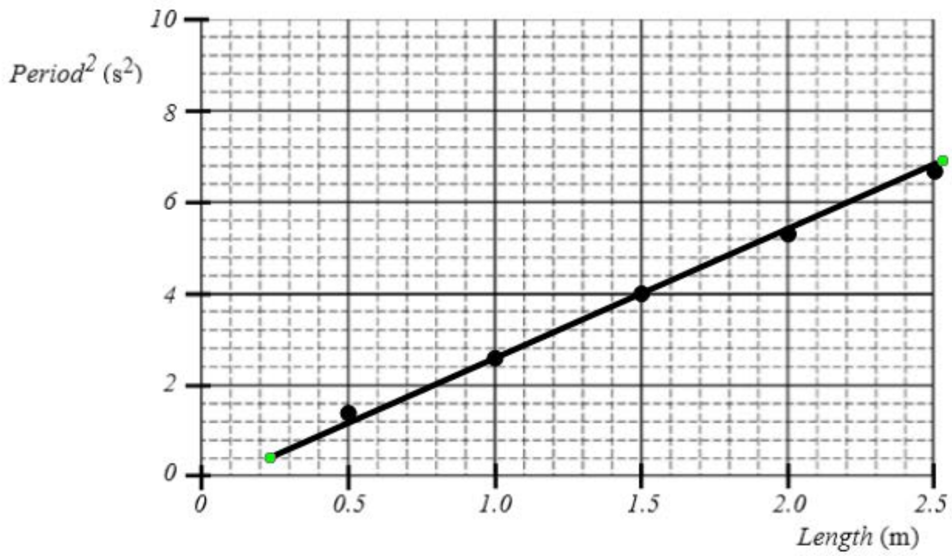
Vertical axis: Period Squared

Note: Square root of Length and Period is also acceptable as is reversing the axes

Question 3 (continued)

Distribution of points

(d) 4 points



For labeling both axes with appropriate variables and units	1 point
For a correct scale that uses more than half the grid	1 point
For correctly plotting data indicated in part (c)	1 point
For drawing a straight line consistent with the given data	1 point

(e) 3 points

For correctly calculating slope using the best-fit straight line and not data points — 1 point

$$\text{slope} = \frac{(y_2 - y_1)}{(x_2 - x_1)} = \frac{(6.0 - 2.0)}{(2.20 - 0.80)} = 2.86\ \text{s}^2/\text{m}$$

Note: linear regression gives $\text{slope} = 2.67\ \text{s}^2/\text{m}$

For correctly relating g to the slope — 1 point

$$T = 2\pi\sqrt{\frac{2L}{3g}} \quad T^2 = \frac{(4\pi^2)2L}{3g} \quad slope = \frac{(4\pi^2)\times 2}{3g}$$

$$g = \frac{8\pi^2}{3 \times slope} = \frac{8\pi^2}{3 \times (2.86\ \text{s}^2/\text{m})}$$

For a correct answer — 1 point

$g = 9.20\ \text{m/s}^2$

Note: linear regression gives $g = 9.86\ \text{m/s}^2$